WALTER AC-PANGAN

CONTACT INFORMATION

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SUMMARY

Agricultural economist with five years of experience at the intersection of forecasting, agribusiness and trade policy. Builds forecasting models with machine learning, remote sensing and econometrics using large field-, county-, state- and country-level spatial-temporal and time series datasets. Experience with causal-inference economic analysis across food supply chains for corn, soybeans, rice, cocoa, beef cattle and citrus in the United States and Latin America. Skilled in several statistical methods, programming languages and coding environments.

ACADEMIC POSITIONS

Virginia Tech
Agribusiness Instructor, Dept. Agricultural and Applied Economics — 2025–2026

Kansas State University
Graduate Research Assistant, Dept. Agricultural Economics — 2019–2025

University of Florida
Research Scholar, Dept. Horticultural Sciences — 2018

University of Minnesota
Visiting Scholar, Dept. Applied Economics — 2017

OTHER PROFESSIONAL EMPLOYMENT

Nuyaak S.A., Guatemala and El Salvador
Financial Analyst — 2026

Oro del Pacífico S.A., Guatemala
Production Assistant Manager, Shrimp Farms — 2018

QualiTech, Chaska, Minnesota
Research Assistant, Agricultural Nutrition Department — 2017

EDUCATION

Kansas State University
Ph.D., Agricultural Economics — 2025
Dissertation: *Essays on Forecasting and International Trade in Grain Markets*
Committee Chair: Nathan P. Hendricks
M.Sc., Agricultural Economics — 2021
Thesis: *How Do Changes in Market Fundamentals Affect Hedging in US Live Cattle Markets?*
Committee Chair: Brian K. Coffey

Zamorano University, Pan-American Agricultural School, Honduras
B.Sc., Agribusiness Management, *summa cum laude* — 2017

PUBLICATIONS

5. "Forecasting Grain Supply of Low and Middle-Income Countries," with N.P. Hendricks, Y.A. Zereyesus, and J. Kee. 2026. *Agricultural and Resource Economics Review*. LINK
4. "Impact of a Soil Conditioner Integrated into Fertilization Scheme on Orange and Lemon Seedling Physiological Performances," with L. Rossi, L.M. Hallman, and S.N. Adams. 2020. *Plants* 9(7): 812. LINK
3. "Effects of Soil Salinity on Citrus Rootstock 'US-942' Physiology and Anatomy," with S.N. Adams and

L. Rossi. 2019. *HortScience* 54(5): 787-792. LINK

2. "Special Article: Projecting Long-Term Staple Grain Production for the International Food Security Assessment: A Time Series Cross-Validation Approach," with Y.A. Zereyesus, J. Kee, N.P. Hendricks, L. Cardell, N.J. Nava, J. Jellife, and S. Morgan. 2023. *USDA ERS International Food Security Assessment, 2023-2033*: 38-47. LINK

1. "Plan de Negocios para el Establecimiento de una Exportadora de Cacao Fino de Aroma en Cobán, Alta Verapaz, Guatemala." 2017. B.Sc. Thesis, Zamorano University, Wilson Popenoe Library. LINK

Working Papers

7. "Identifying Gaps in Food Manufacturing Business: Local Drivers and Impediments for Small Agri-Food Businesses," with P. Sharma, J. Kim, H. Peterson, A. Shanoyan, and J. Yu. (*Revised and resubmitted, Applied Economic Perspectives and Policy*.)

6. "A Unified Algorithm for Penalized Likelihoods in Model Selection of Spatial-Temporal Models for Lattice Data," with F. Jo, P. Reyes, and N.P. Hendricks. (*In submission, Journal of the Royal Statistical Society*.)

5. "In-Season US Crop Acreage Forecasting Using Machine Learning," with Nathan P. Hendricks.

4. "Effects of Non-Tariff Trade Barriers in Grain Markets: The Case of the Rice Export Ban Imposed by India," with Nathan P. Hendricks.

3. "In-Season Forecasting of Corn Yields Using Remote Sensing Data," with M. Ortez, Y. Zhang, and M. Cary.

2. "Basis Prediction Error in Live Cattle Markets Using Machine Learning," with Brian K. Coffey.

1. "Cocoa Commodity Market Analysis: Supply, Marketing, and Price Discovery Mechanisms."

Teaching

Virginia Tech

AAEC 2434: Foundations of Agribusiness (58 students)	2026
AAEC 3454: Small Business Management and Entrepreneurship (52 students)	2026
AAEC 3504: Marketing Agricultural Products (80 students)	2025
AAEC 4514: Advanced Agribusiness Marketing (45 students)	2025

Kansas State University

AGEC 115: Decision Tools for Agricultural Economics and Agribusiness (20 students)	2024
AGEC 632: Agribusiness Logistics (22 students)	2021
R Programming Camp for First-Year M.S. and Ph.D. Students (10 and 14 students)	2023, 2024

Extension and Outreach

Program Support

Assistant Extension Coordinator, 76th Annual Kansas Income Tax Institute, Kansas State University. Logistic support for online and in-person sessions (supervised by Dr. Rich Llewelyn). 2024

Service

Faculty Advisor, Undergraduate Curriculum, Department of Agricultural and Applied Economics, Virginia Tech 2025–2026

Faculty Advisor, Agricultural Economics Club and NAMA Chapter, Virginia Tech 2025–2026

Treasurer, Graduate Students Association, Department of Agricultural Economics, Kansas State University 2022

President, Students Association, Department of Agribusiness, Zamorano University 2016–2017

Honors and Awards

First Place, AAEA / USDA-AMS Local Food Economics Data Visualization Challenge, Agricultural and

Applied Economics Association 2024

First Place, Graduate Level Category, International Food and Agribusiness Management Association (IFAMA) Case Study Competition 2023

Graduate Student Council Travel Award, Kansas State University 2023–2025

First Place in Academic Performance, Department of Agribusiness (*summa cum laude*), Zamorano University 2017

Outstanding Academic Performance Award, Zamorano University 2015–2017

OTHER SKILLS *Methods*: Econometrics, machine learning, forecasting, remote sensing, spatial-temporal and time series analysis, event studies, gravity models, multivariate statistics, risk analysis, DID, Propensity Score Matching, Sythetic Control, and Monte Carlo simulations.
Software: R (R Markdown, Shiny), Python (pandas, NumPy, scikit-learn, XGBoost, statsmodels, TensorFlow, Keras), Stata, SAS, @RISK, Power BI, Excel, LaTeX, Quarto, Git/GitHub, SQL, Microsoft Azure Databricks.

LANGUAGES Spanish (Native) · English (Fluent, C1) · Italian (Conversational, B1) · French (Conversational, A2–B1)

PRESENTATIONS "Machine Learning Approaches for In-Season Forecasting of Corn Yield and Acreage," with N.P. Hendricks. Australasian Agricultural and Resource Economics Society (AARES) Annual Conference, Adelaide, Australia, February 2026.

"Effects of Non-Tariff Trade Barriers in Grain Markets: The Case of the Rice Export Ban Imposed by India," with N.P. Hendricks. AARES Annual Conference, Brisbane, Australia, February 2025.

"Drivers of the Establishment of New Local Food Manufacturing Businesses in the Heartland Region," with P. Sharma and J. Kim. USDA-AMS Local Food Economics Data Visualization Challenge, AAEA Annual Meeting, New Orleans, LA, July 2024.

"Identifying the Untapped Potential of the Small Local Food Manufacturing Business in the US," with P. Sharma and J. Kim. Poster, AI and the Future Symposium, Kansas State University, October 2024.

"Using Machine Learning to Forecast Long-Term Grain Supply of Low and Middle-Income Countries," with N.P. Hendricks. Poster, AI and the Future Symposium, Kansas State University, October 2024.

"Global Grain Supply Forecasting," with N.P. Hendricks, Y.A. Zereyesus, J. Kee, S. Morgan, J. Jellife, N. Nava, and L. Cardell. AAEA Annual Meeting, Washington, DC, July 2023.

"Does Trade Openness Affect Global Grain Supply Response?," with Y.A. Zereyesus, J. Kee, S. Morgan, J. Jellife, N. Nava, L. Cardell, and N.P. Hendricks. AAEA Annual Meeting, Washington, DC, July 2023.

"Food Chain Partnership: When the Funding Stops: Sustaining Private Sector-Non-Profit Collaborations," with L. Hobbs, P. Sharma, C. Ofori-Bah, and M. Sharma. Case Study Solution, International Food and Agribusiness Management Association (IFAMA), Christchurch, New Zealand, June 2023.

"Improving Existing Methods of IFSA Supply Forecasting," with N.P. Hendricks. NCCC-134 Conference on Applied Commodity Price Analysis, Forecasting, and Market Risk Management, St. Louis, MO, April 2023.

"A Unified Algorithm for Penalized Likelihoods in Model Selection of Spatial-Temporal Models for Lattice Data," with F. Jo, P. Reyes, and N.P. Hendricks. Poster, Research and the State, Kansas State University, April 2023.

"How Do Changes in Market Fundamentals Affect Hedging in US Live Cattle Markets?," with B.K. Coffey. NCCC-134 Conference on Applied Commodity Price Analysis, Forecasting, and Market Risk Management (online), April 2021.

"Optimal Nutrient Concentrations and Use of Root Growth Enhancers to Improve Citrus Root Health," with

L. Hallman, L. Rossi, and S.N. Adams. Poster, American Society for Horticultural Science (ASHS) Annual Conference, Las Vegas, NV, July 2019.

"Effects of Soil Salinity on Citrus Rootstock 'US-942' Physiology and Anatomy," with S.N. Adams and L. Rossi. Poster, ASHS Annual Conference, Las Vegas, NV, July 2019.

INVITED SEMINARS

"Forecasting Crop Acreage and Yield in the US: Machine Learning Applications," Department of Agribusiness, Fresno State University, Fresno, CA, March 2026.

"Machine Learning Applications for Forecasting in Agriculture in Latin America," Department of Agribusiness Management, Zamorano University, Honduras (online), June 2025.